HEXLEY CHEMICALS

Permit Standard: Electrical Isolation

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1. Purpose and scope

This standard sets out what a request for its permit type must show before an Area Authority may issue the permit, and what follows the decision.

It is used by the Permit Coordinator or Area Authority who opens a review, by the Area Authority who decides it, and by the performing authority who requests the permit, as the Control of Work Policy describes.

Each permit type has its own permit standard. The five permit standards share one layout and many of the same sentences; they differ in what the permit type covers, in the requirements of section 4 and how section 5 reads them, and in the documentation, the writing that follows a decision and the reasons that sections 6 to 9 give. A request is always reviewed under the standard of its own permit type, and under the standard of its linked permit type where it has one; a requirement in one standard never applies to a request of another permit type unless the request is linked to it.

The standard applies to every request for its permit type, whichever channel it comes through.

2. What this permit covers

This standard covers electrical isolation anywhere at Thornmere Works: making equipment dead, and keeping it dead, for the maintenance of motors, switchgear, cables, lighting and small power, and heaters, at any voltage.

An isolation that another job depends on - the agitator drive of a vessel to be entered, the supply to lighting reached from a platform - is linked to that job's permit, and the work must meet both standards.

The permit type is Electrical isolation.

3. Who requests

A request is made by the performing authority - the contractor's supervisor who will lead the job - through the channel the contractor works in: the site portal, the contractor app or scanned paper. The performing authority answers for the request and the documents attached to it.

The request gives the plant area, its hazardous area classification, whether the process unit there is operating or shut down, whether the work is indoors or outdoors, the permit type and any linked permit type, the work activity, the equipment tag, the size of the work party, the planned work hours and the answers its requirements read. The documents attached to it - a method statement, a task risk assessment, a gas test record or a rescue plan - describe the work.

An isolation request gives the system voltage, whether the isolation is proved dead and whether personal locks are applied, and, above the voltage in section 4, whether circuit earths are applied.

4. Requirements

Every requirement below must be Met for an Area Authority to issue the permit. Each requirement is stated once, in one sentence, and a request is screened against all of them, in this order:

1 **Risk assessment.** A task risk assessment for the job is attached to the request.

2 **Proved dead.** The isolation is proved dead at the point of work.

3 **Personal locks.** Every member of the work party has applied a personal lock to the isolation.

4 **Circuit earths.** On a system above 1000 volts, circuit earths are applied at the point of work.

A requirement is never met by a sentence in a statement, and is never waived.

5. Reading the requirements

Each requirement is Met, Not met or Not documented on the request as the Permit Rules read it. The latest revision of a request is the request; an earlier revision is never screened.

- A task risk assessment is the contractor's own assessment of the job's hazards and the controls against them, attached to the request as one of its documents. The requirement is met when the request records it attached, not met when it records that it is not, and not documented when the answer is blank.
- Proving dead is testing at the point of work that the isolated equipment is dead, with a tester proved before and after the test. The requirement is met when the request records the isolation proved dead, not met when it records that it is not, and not documented when the answer is blank.
- Every member of the work party puts a lock of their own on the isolation, or on the lock box that holds its keys, so that nobody can restore the supply while anyone is at work. The requirement is met when the request records the personal locks applied, not met when it records that it is not, and not documented when the answer is blank.
- Circuit earths are a follow-on question, decided by the system voltage, in whole volts as the Permit Rules convert it from kilovolts. At or below the figure the requirement is met whatever else the request says. Above it, it is met with circuit earths applied, not met without them, and not documented when that answer is blank. A blank system voltage is not documented and asks for the voltage alone.

A requirement that is not documented names the answers it needs, so that the performing authority can give them.

A request with a linked permit type is screened under this standard and under the linked type's, and must meet both. The documents never decide a requirement. A gas testing sentence whose reading differs from the request's is flagged for the Area Authority, and the request's answer still decides.

6. Documentation to provide

When a request is returned to its requester, the performing authority must provide this documentation and send the request again with it: a task risk assessment for the job, the isolation certificate with the isolation points and the system voltage, the proving dead test, the personal lock register for the work party, and on a system above the voltage in section 4 the circuit earths record.

7. The permit document

The permit document goes to the performing authority when an Area Authority issues a permit under this standard. It names the request by its permit reference and permit type, says the permit is issued, and gives the plant area, the equipment tag, the work activity, the work party's size and the planned hours, with the system voltage, the proving dead, the personal locks and, above the voltage in section 4, the circuit earths. A linked permit type is named with its own controls. It asks the performing authority to keep every personal lock on until the permit is handed back. The permit document carries no figure from a model.

8. The work party briefing

The work party briefing is written after the permit document, for the people who will do the work. It tells them in plain words what the job is, where it is, its hazards and the controls they must keep. For an electrical isolation it says that everyone puts a lock of their own on the isolation, that the equipment is proved dead before it is touched, and that nobody removes another person's lock. Where the permit depends on a linked permit, the briefing says so. The briefing never gives the suspension likelihood, the expected work hours or any other figure from a model, and never names any member of Hexley's staff.

9. Why these requirements

An isolation is safe only when the equipment is proved dead at the point of work, and stays safe only while nobody can restore the supply, so every member of the work party applies a lock of their own.

On a system above the voltage that section 4 gives, a dead conductor can still hold a dangerous charge, or be made live by induction or by a switching mistake, so circuit earths are applied as well.

Process Safety owns this standard, reviews it every twelve months and issues any amendment as a new issue of this document, replacing every earlier issue.